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ChE 311 Midterm Exam I, Spring 2019

Closed book and closed notes. Only provided tables are allowed.

50 minutes, 100 points

1. (5 points) For the following velocity profile,

$$\vec{v} = (5t + 7x - 3y + z)\vec{\delta}_x + (x - 2y + z)\vec{\delta}_y + (5t + x - y + 5z)\vec{\delta}_z$$

The flow is incompressible.

(A) True

(B) False

(C) Not enough information to decide

$$\nabla \cdot \rho \vec{v}$$

$$\left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle v_x, v_y, v_z \rangle$$

$$(7 + (-2) + 5)$$

10

2. (5 points) The flow defined by the following velocity profile with a cylindrical coordinate satisfies the continuity equation. For the flow, density is constant.

$$\vec{v} = (-5r)\vec{\delta}_r + (2t + r + 5z)\vec{\delta}_\theta + (5z)\vec{\delta}_z$$

(A) True

(B) False

(C) Not enough information to decide

$$\frac{1}{r} \frac{\partial}{\partial r} (r(-5r)) + \frac{1}{r} \left( \frac{\partial}{\partial \theta} 0 \right) + \frac{\partial}{\partial z} (5z)$$

$$\frac{1}{r} \frac{\partial}{\partial r} (-5r^2)$$

$$\frac{1}{r} (-10r)$$

$$-10$$

$$0 \quad 5$$

$$= -5$$

3. (10 points) Please choose the answer for the following questions.

$$\vec{u} = (3x_1 + 4x_2)\vec{\delta}_1 + (x_2 + 4x_3)\vec{\delta}_2 + (x_1 + 2x_3)\vec{\delta}_3$$

(1)  $\nabla \cdot \vec{u} =$

(a) 16

(b) 6

(c) 10

(d) 5

(2)  $\nabla \vec{u} =$

(a)  $\begin{vmatrix} 3 & 4 & 1 \\ 1 & 1 & 4 \\ 1 & 1 & 2 \end{vmatrix}$

(b)  $\begin{vmatrix} 3 & 4 & 0 \\ 0 & 1 & 4 \\ 1 & 0 & 2 \end{vmatrix}$

(c)  $\begin{vmatrix} 3 & 1 & 1 \\ 4 & 1 & 1 \\ 1 & 4 & 2 \end{vmatrix}$

(d)  $\begin{vmatrix} 3 & 0 & 1 \\ 4 & 1 & 0 \\ 0 & 4 & 2 \end{vmatrix}$

$$\left( \sum_i \delta_i \frac{\partial}{\partial x_i} \right) \left( \sum_j \delta_j V_j \right)$$

$$\sum_i \delta_i \delta_j \frac{\partial V_j}{\partial x_i}$$

$$\sum_i \delta_i \delta_j \frac{\partial}{\partial x_i} V_j$$

$$\delta_1 \delta_1 \quad \delta_1 \delta_2 \quad \delta_1 \delta_3$$

$$\delta_2 \delta_1 \quad \delta_2 \delta_2 \quad \delta_2 \delta_3$$

$$\delta_3 \delta_1 \quad \delta_3 \delta_2 \quad \delta_3 \delta_3$$

4. (5 points)

$$\vec{\tau} = \begin{vmatrix} 2x_1 + x_2 & x_2 + x_3 & 3x_1 + x_3 \\ x_1 & x_2 & x_3 \\ 0 & x_3 & x_1 \end{vmatrix}$$

Please choose the right answer for  $\nabla \cdot \vec{\tau}$

(a)  $\nabla \cdot \vec{\tau} = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 4 \\ 0 & 1 & 0 \end{vmatrix}$

(b)  $\nabla \cdot \vec{\tau} = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 4 & 0 \end{vmatrix}$

(c)  $\nabla \cdot \vec{\tau} = 3\delta_1 + \delta_2 + 3\delta_3$  (d)  $\nabla \cdot \vec{\tau} = 4\delta_1 + 3\delta_2 + \delta_3$

$$\sum_i \delta_i \frac{\partial}{\partial x_i} \cdot \sum_j \delta_j \frac{\partial}{\partial x_j}$$

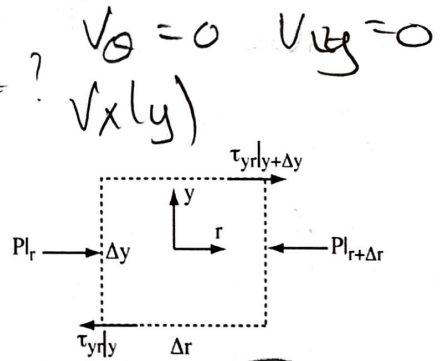
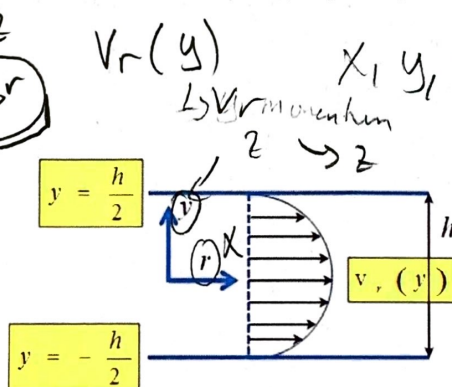
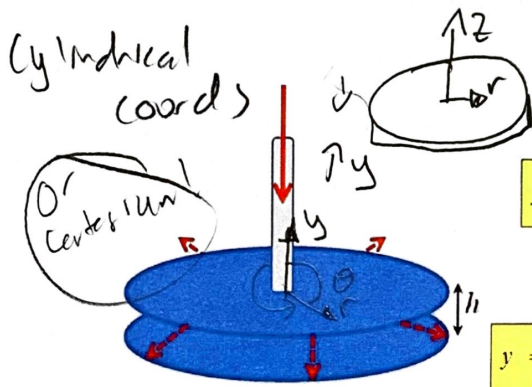
$$\sum_{ijk} \delta_i \delta_j \delta_k \frac{\partial \tau_{ijk}}{\partial x_i}$$

$$\delta_{ij} = 1 \text{ at } i=j$$

$$\Rightarrow \sum_{ijk} \delta_k \frac{\partial \tau_{ijk}}{\partial x_i}$$

5. A radial-flow channel has been developed to measure the adhesion (energy) of cells to surfaces *in vitro*. The radial-flow channel is shown in the figure below. In this device, fluid flows in from a central source, flows out in a radial direction and the volumetric flow rate at any positions  $r$  is  $Q$ . The flow itself is driven by an unspecified pressure gradient.

- (1) Solve for the velocity profile (20 points).
- (2) Use your solution for the velocity profile and the constant flow rate condition to calculate the wall shear stress at  $r = 1$  cm. The fluid viscosity is similar as water  $1 \text{ mPa}\cdot\text{s}$ . The volumetric flow rate is  $10 \mu\text{L}/\text{min}$ , with  $h = 200 \mu\text{m}$ . (20 points)



$\rightarrow 0 = -\frac{\partial P}{\partial r} + \mu \left( \frac{\partial^2 v_r}{\partial y^2} \right) + \rho g_r$

$\rightarrow 0 = \frac{1}{r} \frac{\partial P}{\partial \theta} \Rightarrow P \neq f(\theta)$

$\rightarrow 0 = -\frac{\partial P}{\partial z} + 0 + \rho g_z \Rightarrow P \neq f(z)$

$\frac{dP}{dr} = \mu \frac{\partial^2 v_r}{\partial y^2}$

$\Rightarrow \frac{dP}{dr} = \mu \frac{d^2 v_r}{dy^2} = C_0$

$\frac{\partial P}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (v_\theta v_\theta) + \frac{\partial}{\partial z} (v_z v_z) = 0$

$$\frac{dP}{dr} = \mu \frac{d^2 v_r}{dy^2} = C_0$$

$$\Rightarrow \frac{dP}{dr} = C_0$$

$$\int_{P_1}^{P_2} dP = \int_0^R C_0 dr$$

$$P_2 - P_1 = C_0 R - 0$$

$$C_0 = \frac{P_2 - P_1}{R}$$

$$\mu \frac{d^2 v_r}{dy^2} = C_0$$

$$\frac{d^2 v_r}{dy^2} = \frac{C_0}{\mu}$$

$$\int d \frac{d^2 v_r}{dy^2} = \int \frac{C_0}{\mu} dy$$

$$\frac{dv_r}{dy} = \frac{C_0}{\mu} y + C_1$$

$$\int dv_r = \int \frac{C_0}{\mu} y dy + \int C_1 dy$$

$$v_r = \frac{C_0}{2\mu} y^2 + C_1 y + C_2 \quad \checkmark$$

BC 1

$$y = 0$$

$$\frac{dv_r}{dy} = 0$$

BC 2

$$y = \frac{h}{2}$$

$$v_r = 0 \quad \checkmark$$

$$0 = 0 + C_1$$

$$C_1 = 0$$

$$C_2 = \frac{C_0}{2\mu} \left(\frac{h}{2}\right)^2$$

Then ?  $\rightarrow v_r = \frac{C_0}{2\mu} y^2$



6. You need to design a piece of pipe flow for a microfluidic system. Consider the flow is at steady state. Neglect the "end effects". The liquid is a Newtonian fluid with constant density and viscosity.

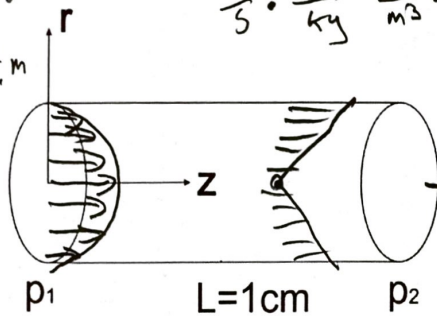
$\nu = 2 \times 10^{-4} \text{ m}^2/\text{s}$  is the kinematic viscosity of the fluid. The density of the fluid is  $\rho = 0.8 \times 10^3 \text{ kg/m}^3$ .

The circular pipe is placed horizontal (figure 1). The length of the pipe is 1.5 cm. Flow is from left to right and driven by a syringe pump. The pressure drop for this piece of pipe is  $P_1 - P_2 = 0.25 \text{ N/m}^2$ . (hint:

$P = P(z)$ ). The special material used to make this pipe can only afford the maximum shear stress  $0.02 \text{ N/m}^2$  before yielding.

(1) With above yield stress, what is the maximum radius of the tube,  $R$ ? (25 points)

(2) What is the maximum velocity? (10 points)



$$v_\theta = 0 \quad v_r = 0$$

$$\text{cont. } \frac{\partial p}{\partial z} + 0 + 0 \quad \frac{\partial}{\partial z}(\rho v_z) = 0$$

$$v_z \neq f(z)$$

z-mom. N-S

$$\Rightarrow 0 = -\frac{\partial p}{\partial z} + \mu \left( \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial v_z}{\partial r} \right) \right) + \cancel{\rho g_z}$$

$$\frac{\partial p}{\partial z} = \frac{\mu}{r} \left( \frac{\partial}{\partial r} \left( r \frac{\partial v_z}{\partial r} \right) \right)$$

$$p = P(z)$$

$$v_z = V(r) \Rightarrow \frac{dp}{dz} = \frac{\mu}{r} \frac{d}{dr} \left( r \frac{dv_z}{dr} \right) = C_0 \quad \checkmark$$

$$\Rightarrow \frac{dp}{dz} = C_0$$

$$\int_{p_1}^{p_2} dp = \int_0^L C_0 dz$$

$$p_2 - p_1 = C_0 z \Big|_0^L$$

$$p_2 - p_1 = C_0 L - 0$$

$$\Rightarrow C_0 = \frac{p_2 - p_1}{L}$$

$$\Rightarrow \frac{\mu}{r} \frac{d}{dr} \left( r \frac{dv_z}{dr} \right) = C_0$$

$$\int d \left( r \frac{dv_z}{dr} \right) = \int \frac{C_0}{\mu} r dr$$

$$r \frac{dv_z}{dr} = \frac{C_0}{2\mu} r^2 + C_1$$

$$\frac{dv_z}{dr} = \frac{C_0}{2\mu} r + \frac{C_1}{r}$$

$$\int dv_z = \int \left( \frac{C_0}{2\mu} r dr + \right) \frac{C_1}{r} dr$$

$$v_z = \frac{C_0}{4\mu} r^2 + C_1 \ln(r) + C_2 \quad \checkmark$$

$$C_0 = \frac{P_2 - P_1}{L}$$

$$V_z = \frac{C_0}{4\mu} r^2 + C_1 \ln(r) + C_2 \quad \frac{dV_z}{dr} = \frac{C_0}{2\mu} r + \frac{C_1}{r}$$

$$\text{BC1} \quad \frac{dV_z}{dr} \sim \text{finite} \quad r=0 \quad \text{BC2} \quad V_z = 0 \quad r=R$$

$$0 = 0 + \frac{C_1}{0}$$

$$C_1 = 0$$

$$\rightarrow 0 = \frac{C_0}{4\mu} R^2 + 0 + C_2$$

$$C_2 = -\frac{C_0}{4\mu} R^2 \quad \checkmark$$

$$\Rightarrow V_z = \frac{C_0}{4\mu} r^2 - \frac{C_0}{4\mu} R^2$$

$$V_z = \frac{(P_2 - P_1)}{4\mu L} r^2 - \frac{P_2 - P_1}{4\mu L} R^2 \quad \checkmark$$

$$\tau_{rz} = -\mu \left( 0 + \frac{\partial V_z}{\partial r} \right)$$

$$= -\mu \frac{\partial}{\partial r} \left( \frac{C_0}{4\mu} r^2 - \frac{C_0}{4\mu} R^2 \right)$$

$$= -\mu \left[ \frac{C_0}{2\mu} r - 0 \right]$$

$$\tau_{rz} = -\frac{C_0}{2} r$$

$$r = \frac{2\tau_{rz}}{C_0} = \frac{2\tau_{rz}}{\frac{P_1 - P_2}{L}}$$

$$r = \frac{2 \cdot (0.02 \text{ N/m}^2)}{\frac{0.25 \text{ N/m}^2}{1 \text{ cm}}}$$

$$r = 0.16 \text{ cm} \quad \checkmark$$

$$V_{z \max} \Rightarrow r=0$$

$$V_z = 0 - \frac{C_0}{4\mu} R^2$$

$$= -\frac{\frac{P_1 - P_2}{L}}{4\mu} R^2$$

$$\mu = \rho \nu$$

$$\frac{0.25 \text{ N/m}^2}{1 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}}} \cdot (0.16 \text{ cm})^2$$

$$\frac{4(0.8 \times 10^3 \text{ kg/m}^3)(2 \times 10^{-4} \text{ m}^2/\text{s})}{25 \text{ N/m}^2} \cdot 2.25 \times 10^{-6}$$

$$= \frac{0.64 \text{ kg}}{\text{ms}}$$

$$= 8.789 \times 10^{-5} \frac{\text{ms}}{\text{kg}} \cdot \frac{\text{kg}}{\text{ms}} \cdot \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$V_{\max} = 8.789 \times 10^{-5} \text{ m/s}$$